

Data Visualisation

Day 1 — From Data to Insight

Course Introduction · Data Fundamentals · Tools · Data Types

9-week programme · Excel · Python

Agenda — Day 1

Morning

1. Our tools
2. Data → Information → Insight
3. Why visualisation matters
4. The visualisation workflow

Afternoon

5. Data types & why they matter
6. How to choose the right chart
7. How to make an effective chart
8. Hands-on exercise
9. Q & A + homework

By end of day you will understand **what data is**, **what it becomes**, and **how we work with it** for the rest of this course.

01

Our Tools

Excel · Python · *maybe* AI

Three Tools, One Workflow

Microsoft Excel

- Business reports
- Quick prototypes
- PivotTables
- One-off analysis

Insert → **Charts wizard**

Python

- Large datasets
- Automation & scripts
- Publication quality
- Reproducible analysis

`matplotlib`

`seaborn`

`plotly`

*Knowing **which tool to reach for** is itself a professional skill.*

Excel — Core Skills We Will Cover

Data work

- PivotTable — group, aggregate, pivot
- VLOOKUP / XLOOKUP — merge tables
- Conditional formatting
- Data validation

Charts

Line · Bar · Waterfall

Funnel · Stock (Candlestick) · Combo

SALES BY MONTH

Monthly sales in *mil* EUR

	2019	2020	%pct
Jan	4.5	5.2	15%
Feb	2.9	2.0	-32%
Mar	2.5	3.6	42%
Apr	4.2	4.1	-2%
May	3.8	5.5	45%
Jun	7.7	7.2	-7%
Jul	6.6	5.2	-21%
Aug	3.4	4.7	37%
Sep	4.9	3.8	-22%
Oct	4.4	4.9	11%
Nov	3.0	4.7	60%
Dec	8.7	5.9	-32%
	56.6	56.7	0.2%

Python – Three Libraries, Three Purposes

matplotlib

The **foundation**. Full manual control over every element.

```
import matplotlib.pyplot as plt

plt.plot(months, revenue,
         color='#0D1A63')
plt.title('Monthly Revenue')
plt.show()
```

seaborn

Statistical charts with minimal code. Built on matplotlib.

```
import seaborn as sns

sns.boxplot(
    x='Group',
    y='Score',
    data=df
)
```

plotly

Interactive charts — hover, zoom, filter. Web & dashboards.

```
import plotly.express as px

fig = px.bar(df,
             x='Month',
             y='Revenue')
fig.show()
```

02

What Is Data?

And Why Does It Matter?

Data · Information · Insight

A single transaction:

Invoice

Craigs' Landscaping
123 Sierra Dr
Palo Alto, CA 94028

123-456-7890
craigslandscaping@email.com
craigslandscaping.com

Bill To

Smith Enterprises
456 Sierra Dr
Palo Alto, CA 94028

Invoice Details

Invoice # 12345
Terms Net 30
Invoice date 08/01/2024
Due date 08/31/2024

Product/Services	Qty/Hrs	Rate	Amount
Stone work	3.4	\$125	\$425
Mulching, weeding, planting	3	\$75	\$225
Subtotal			\$650.00
Discount			-\$65.00
Tax 8.5%			\$49.73
Total			\$634.73

Table record each transaction:

Date	Invoice	Customer	Product	Qty	Unit Price	Total
2024-01-03	INV-001	Alice Johnson	Laptop X1	1	\$899	\$899
2024-02-14	INV-002	Bob Martinez	Mouse Pro	3	\$120	\$360
2024-03-07	INV-003	Carol White	Laptop X1	1	\$899	\$899
2024-01-22	INV-004	David Lee	Keyboard K2	2	\$150	\$300
2024-04-11	INV-005	Emma Brown	Monitor 27"	1	\$450	\$450
2024-05-03	INV-006	Frank Wilson	Mouse Pro	2	\$120	\$240
2024-02-28	INV-007	Grace Taylor	Laptop X1	1	\$899	\$899
2024-06-15	INV-008	Henry Clark	Monitor 27"	2	\$450	\$900
2024-03-19	INV-009	Irene Scott	Keyboard K2	1	\$150	\$150
2024-05-27	INV-010	James Rivera	Mouse Pro	5	\$120	\$600
...						

Data · Information · Insight

Table record each transaction:

Date	Invoice	Customer	Product	Qty	Unit Price	Total
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2024-01-22	INV-004	David Lee	Keyboard K2	2	\$150	\$300
2024-04-11	INV-005	Emma Brown	Monitor 27"	1	\$450	\$450
2024-05-03	INV-006	Frank Wilson	Mouse Pro	2	\$120	\$240
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...						

Sum of transactions by month:

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Data · Information · Insight

Data

Raw, unprocessed facts — no context, no meaning.

42, 87, 13, 56

2024-01-03, North, 5000

Information

Data **with context** — structured, labelled, comparable.

"Monthly revenue Jan–Apr: 42k, 87k, 13k, 56k USD"

Insight

Information that drives **action** — the "so what?" answer.

"Revenue collapsed in March — investigate supply chain."

Visualisation is the bridge that turns information into insight at a glance.

Why Visualisation Matters

Without a chart:

What insight can you see from the data?

SALES BY MONTH

Monthly sales in *mil* EUR

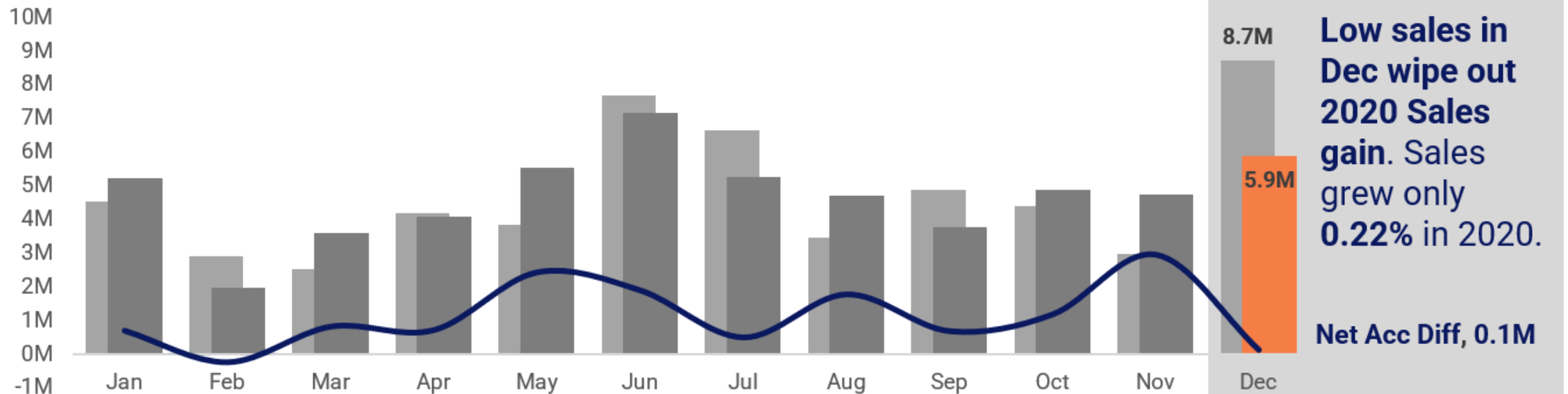
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Why Visualisation Matters

With a chart:

LOSS OF CHRISTMAS SALES

Monthly Sales 2019 vs 2020 (EUR)



The Same Numbers – Three Levels

Level	What you see	What you can do
Data	42, 55, 48, 62, 78, 85, 92, 88, 95, 102, 98, 115	Nothing yet
Information	Monthly revenue Jan–Dec (USD thousands)	Begin comparing
Insight	Revenue grew 174% over the year; Q3 was the inflection point	Make a decision

The Visualisation Workflow

Every session in this course follows the same pipeline:



```
graph LR; A[Raw Data] --> B[Clean & Structure]; B --> C[Presentation Table]; C --> D[Chart / Dashboard]; D --> E[Story + Decision]
```

Raw Data ▶ Clean & Structure ▶ Presentation Table ▶ Chart / Dashboard ▶ Story + Decision

We cover **every step** — not just the chart at the end.

The table step is where most analysts lose quality.

Ex01.01

Raw Data ▶ Clean & Structure ▶ Presentation Table

Ronnykym. (2020). Online Store Sales Data [Data set]. Kaggle.

Use Excel formulas or pivot table to report on sale by month of the year 2019 and 2020.



Ex01.01

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03

Data Types

An Important Concept in Visualisation

Quantitative — Continuous vs Discrete

Continuous

Can take **any value** in a range — infinitely many possible values exist between two points.

Examples

- Revenue: \$47,382.56
- Temperature: 36.8°C
- Height: 1.73 m
- Profit margin: 12.4%

Key charts

Line · Scatter · Histogram · Box Plot

Discrete

Countable whole numbers only.
You cannot have 2.5 customers.

Examples

- Customers: 847
- Units sold: 1,200
- Page views: 54,321
- Support tickets: 7

Key charts

Bar · Column · Dot Plot

Categorical — Nominal vs Ordinal

Nominal

Categories with **no natural order**.

You cannot say North > South.

Examples

- Region: North / South / East / West
- Product: Widget A / Widget B
- Channel: Email / Social / Search
- Country: Vietnam / USA / UK

Key charts

Bar · Donut · Treemap

Ordinal

Categories **with a meaningful order** — but the gap between levels may not be equal.

Examples

- Rating: Poor · Fair · Good · Excellent
- Education: High School · Bachelor's · Master's
- Priority: Low · Medium · High · Critical
- NPS: Detractor · Passive · Promoter

Key charts

Bar (sorted) · Heatmap · Stacked Bar

Temporal – Time-based Data

Data recorded at **points in time** or over **time intervals**.

Regular intervals

Daily stock prices, monthly sales, hourly temperature readings.

The gap between points is constant — patterns like seasonality and trends become visible.

Charts: Line · Area · Candlestick · Calendar Heatmap

Irregular intervals

Customer purchase timestamps, server log events, support tickets.

Points are unevenly spaced — aggregation (hourly → daily → monthly) is needed first.

Charts: Line after aggregation · Event timeline

Golden rule: Time always goes on the **X-axis**. The metric goes on the **Y-axis**.

Geospatial Data

Data with a **location** component.

Types of location data

Type	Example
Country name	"Vietnam", "USA"
Region / Province	"North", "HCMC"
Coordinates	lat 10.8°, lon 106.7°
Postal code	70000
Address	"123 Nguyen Hue St"

Key charts

Choropleth map — colour regions by value (sales by country)

Dot / bubble map — sized dots at coordinates (store locations)

Geo heatmap — density of events (customer clusters)

We cover choropleth maps using `plotly` in the Tools section.

Why Data Types Matter

The data type determines the chart — not the other way around.

Trying to draw a line chart on nominal categories, or a donut chart on continuous data, produces charts that are meaningless — or actively misleading.

Data type	Examples	Suitable charts
Continuous	Revenue, temperature, height	Line, scatter, histogram
Discrete	Customers, units, page views	Bar, dot plot
Nominal	Country, product name, channel	Bar, treemap
Ordinal	Rating: Poor / Good / Excellent	Bar, heatmap
Temporal	Dates, timestamps	Line, area, candlestick
Geospatial	City, coordinates, country	Map, choropleth

Ex01.02

Identify the data type of each column in the Ex01 dataset.

For each column, classify as: Continuous · Discrete · Nominal · Ordinal · Temporal · Geospatial

Sample Data from Ex01

country	order_value	cost	date	category	customer_name	sales_manager	sales_rep	device_type	order_id
Sweden	17,524.02	14,122.61	2020-02-12	Books	Goldner-Dibbert	Maxie Marrow	Madelon Bront	Mobile	70-0511466
Finland	116,563.40	92,807.78	2019-09-26	Games	Hilli-Vandervort	Hube Corey	Wat Bowkley	Mobile	28-6585323
Portugal	296,465.56	257,480.34	2019-07-11	Clothing	Larkin-Collier	Celine Tumasian	Smitty Culverhouse	PC	58-7703341
Portugal	74,532.02	59,752.32	2020-04-02	Beauty	Hessel-Stiedemann	Celine Tumasian	Aurelie Wren	PC	14-6700183
Spain	178,763.42	146,621.76	2019-12-22	Games	Johns and Sons	Emalia Dinse	Bertha Walbrook	Tablet	15-8765160
Portugal	71,620.08	62,245.01	2019-02-05	Books	Schoen-Keeling	Celine Tumasian	Smitty Culverhouse	PC	69-6259390
Portugal	78,461.13	63,537.82	2020-05-10	Appliances	Hessel-Stiedemann	Celine Tumasian	Smitty Culverhouse	Mobile	91-4126746
France	64,827.80	56,043.63	2019-01-20	Appliances	Gislason-Stanton	Othello Bowes	Maighdiln Upcraft	PC	62-3312495
UK	66,673.19	52,811.83	2019-02-23	Smartphones	Gislason-Stanton	Jessamine Apark	Winny Agnolo	PC	25-6368157
Portugal	136,915.61	114,790.05	2020-01-04	Outdoors	Hessel-Stiedemann	Celine Tumasian	Aurelie Wren	Tablet	15-3943365

Data Dictionary and Types

Column	Description	Data Type
country	Customer Origin Country	Categorical - Nominal
order_value_EUR	Order_value	Continuous
cost	Cost of goods sold	Continuous
date	date ordered	Temporal
category	Product Category	Categorical - Nominal
customer_name	Customer Name	Categorical - Nominal
sales_manager	Sale Manager Name	Categorical - Nominal
sales_rep	Sale Representative Name	Categorical - Nominal
device_type	The device type customer used to order - Nominal	Categorical - Nominal
order_id	unique order_id	Categorical - Nominal

04

How to Choose the Right Chart

Abela's Framework · FT Visual Vocabulary · Domain Examples

Two Frameworks for Chart Selection

*There are many different frameworks. These just serve as guides and are **not rules written in stone***

Abela's Decision Tree

Start with **what you want to show** — Comparison, Distribution, Relationship, or Composition — then follow the branches to the right chart.

Developed by Andrew Abela (2006). Works best as a **quick first filter** when you already know your data type.

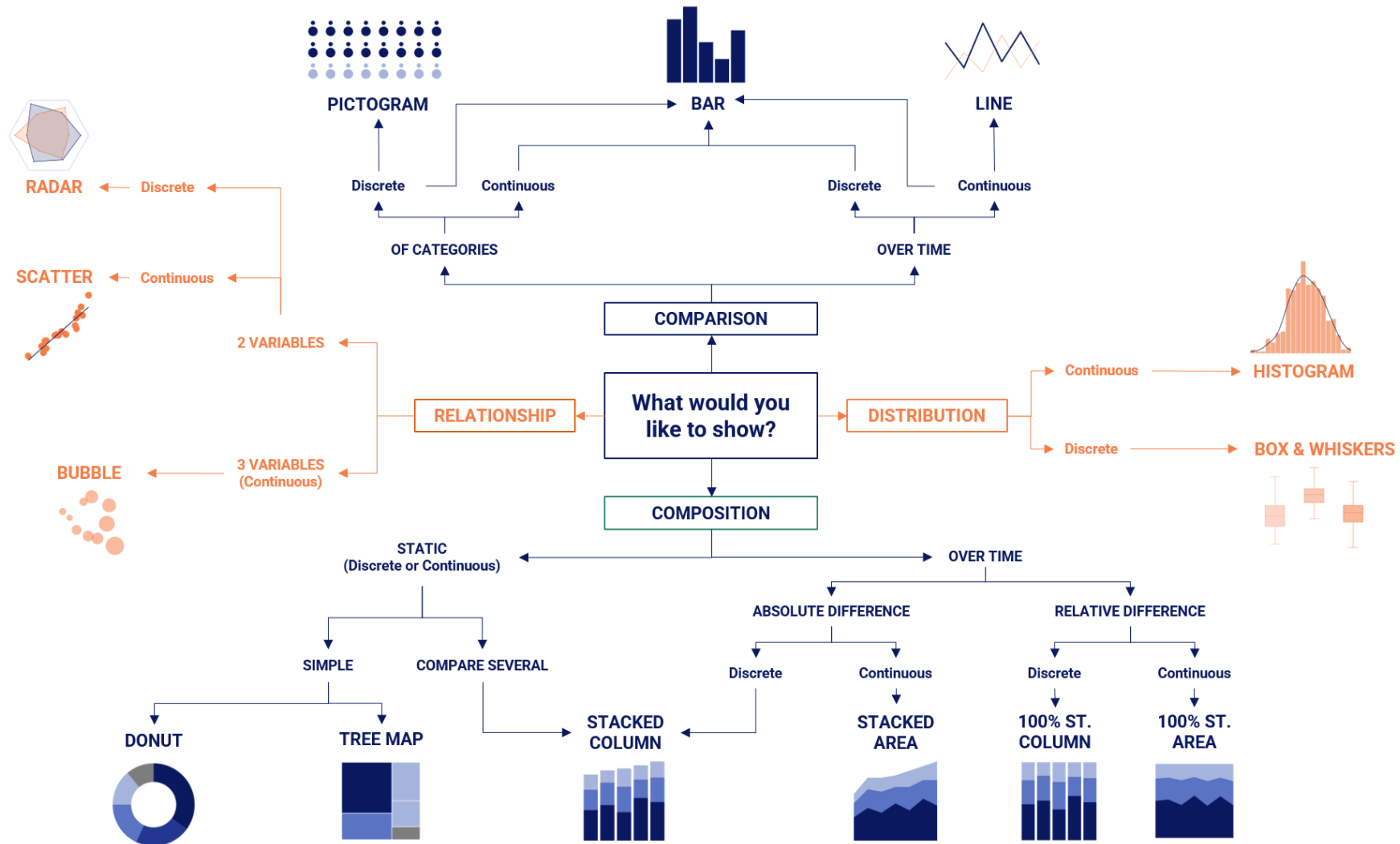
FT Visual Vocabulary

Organised by **analytical purpose** across 9 families — Deviation, Correlation, Change V Time, Ranking, Distribution, Part of Whole, Magnitude, Spatial, Flow.

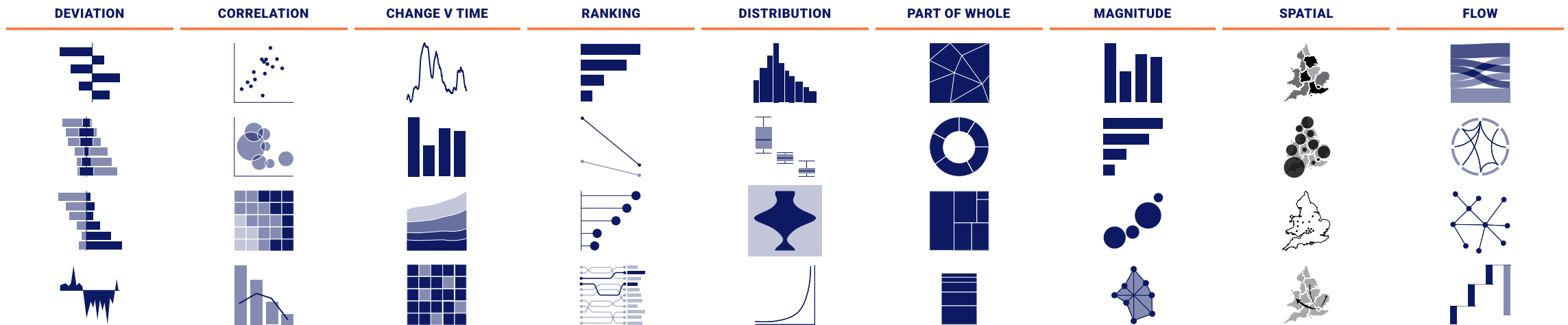
Developed by the Financial Times visual journalism team. Covers **59 chart types** with real-world editorial context.

Both frameworks start with the same question: "**What does your data need to say?**" — then map it to the right visual form.

The Data-Type Decision Tree



Financial Times Visual Vocabulary

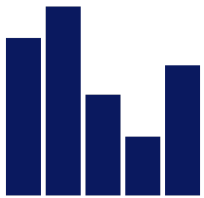


Financial Times. (n.d.). *Visual vocabulary*. <https://ft-interactive.github.io/visual-vocabulary/>

Chart Types — By Purpose

Comparison

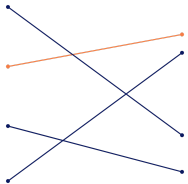
BAR



LINE

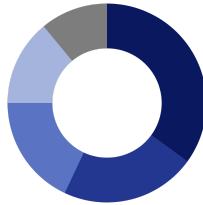


SLOPE

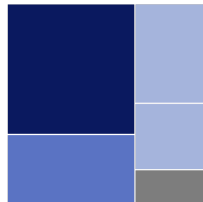


Composition

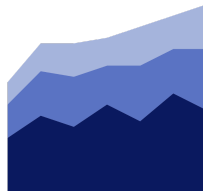
DONUT



TREEMAP

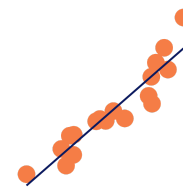


STACKED AREA

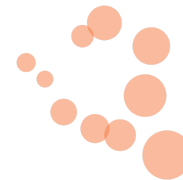


Relationship

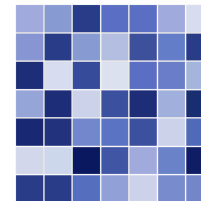
SCATTER



BUBBLE

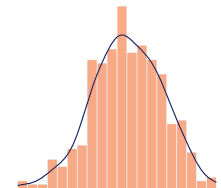


HEATMAP

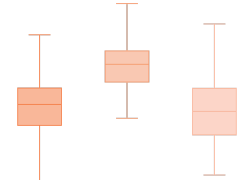


Distribution

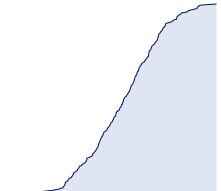
HISTOGRAM



BOX PLOT

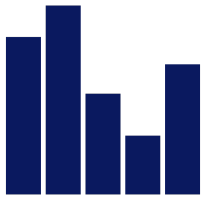


CDF



Finance Charts — Trends & Composition

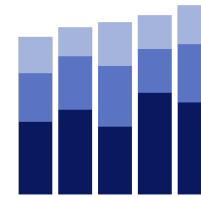
BAR



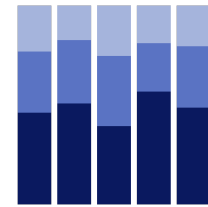
LINE



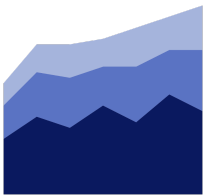
STACKED COLUMN



100% STACKED COLUMN



STACKED AREA



100% STACKED AREA



AREA

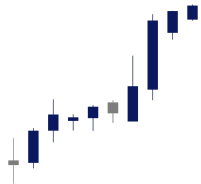


WATERFALL



Finance Charts – KPI, Prices & Allocation

CANDLESTICK – OHLC STOCK PRICES



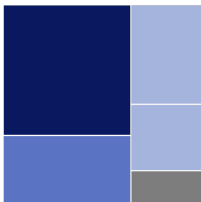
BULLET – KPI VS TARGET



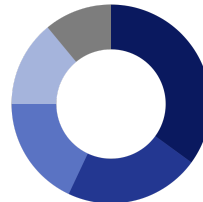
GAUGE – SINGLE KPI DIAL



TREEMAP – PORTFOLIO / BUDGET BREAKDOWN

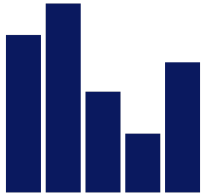


DONUT – ASSET / REVENUE MIX



Sales & Marketing Charts — Comparison & Composition

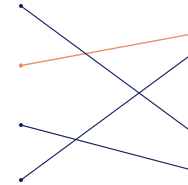
BAR



LINE



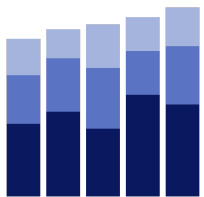
SLOPE



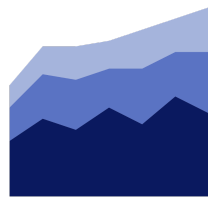
AREA



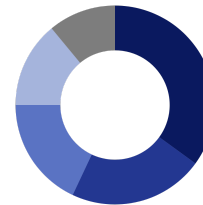
STACKED COLUMN



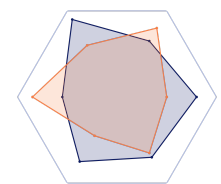
STACKED AREA



DONUT



RADAR

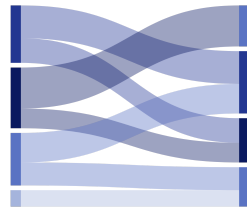


Sales & Marketing Charts — Flow, Pipeline & Insights

FUNNEL



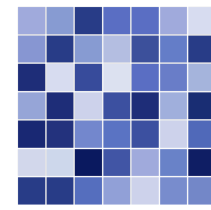
ALLUVIAL



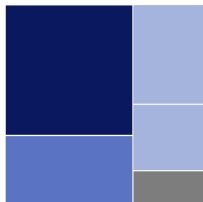
SANKEY



HEATMAP



TREEMAP

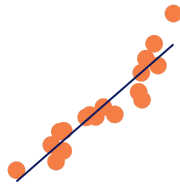


PICTOGRAM

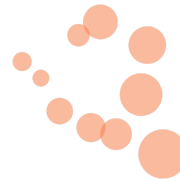


Science & Research Charts – Relationships & Patterns

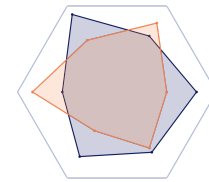
SCATTER – CORRELATION (2 VARIABLES)



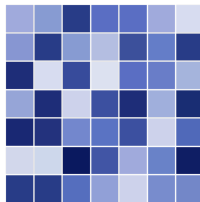
BUBBLE – 3-VARIABLE RELATIONSHIP



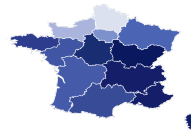
RADAR – MULTI-ATTRIBUTE PROFILE



HEATMAP – PATTERNS ACROSS TWO DIMENSIONS



CHOROPLETH – METRIC BY REGION

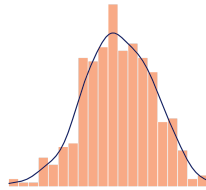


LINE – TIME-SERIES TRACKING

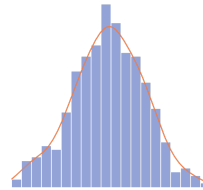


Science & Research Charts – Distributions

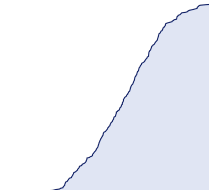
HISTOGRAM – FREQUENCY DISTRIBUTION



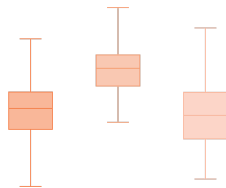
HISTOGRAM + KDE – WITH DENSITY CURVE



CDF – CUMULATIVE PROBABILITY



BOX PLOT – MEDIAN & SPREAD



Ex01.03 – Formulating Questions

Activity: What can we ask the data?

Take a moment to look at the dataset structure and data types.

What are some analytical questions we can answer using this data?



20 Analytical Questions for Ex01.03

With our dataset mapped out, what can we ask it? Here are 20 questions we can answer using visualization:

1. What is the total overall revenue and total cost?
2. Which **country** generated the highest total revenue?
3. What is the average order value across all transactions?
4. How do sales trend over time (**date**) on a monthly basis?
5. Which product **category** yields the highest total profit?
6. Who are the top 5 customers (**customer_name**) by total volume?
7. Which **sales_manager** has the highest-performing team?
8. Which **sales_rep** closed the highest number of orders?
9. Are there noticeable seasonal spikes throughout the year?
10. What is the most frequently used **device_type** for ordering?

20 Analytical Questions (cont.)

11. Is there a correlation between device and average order value?
12. Which country has the lowest average profit margin per order?
13. How does the product mix (**category**) differ by country?
14. Which customer has the highest single order value?
15. What percentage of revenue comes from the top category?
16. Does the time of year influence category popularity?
17. Do specific reps specialize in high-value vs. volume orders?
18. How does total cost vary relative to revenue across countries?
19. Which device and country combo yields the highest order value?
20. What is the distribution of order values (small vs large)?

Ex01.04 – Chart Selection Activity

Activity: Choose the Right Chart

1. One student selects a question from the 20 analytical questions.
2. The class discusses and decides **what chart type** is best to answer it.
3. Justify your choice based on the data types and the message you want to convey.

05

Chart Journey

From Table to Insight — A Practical Walkthrough

Step 0 – The Presentation Table

Before drawing any chart, you need a **clean presentation table** – aggregated, labelled, and ready to visualise.

This is the output of **Ex01.01**: monthly sales by year, condensed from thousands of raw transactions into 24 numbers.

SALES BY MONTH

Monthly sales in *mil* EUR

	2019	2020	%pct
Jan	4.5	5.2	15%
Feb	2.9	2.0	-32%
Mar	2.5	3.6	42%
Apr	4.2	4.1	-2%
May	3.8	5.5	45%
Jun	7.7	7.2	-7%
Jul	6.6	5.2	-21%
Aug	3.4	4.7	37%
Sep	4.9	3.8	-22%
Oct	4.4	4.9	11%
Nov	3.0	4.7	60%
Dec	8.7	5.9	-32%
	56.6	56.7	0.2%

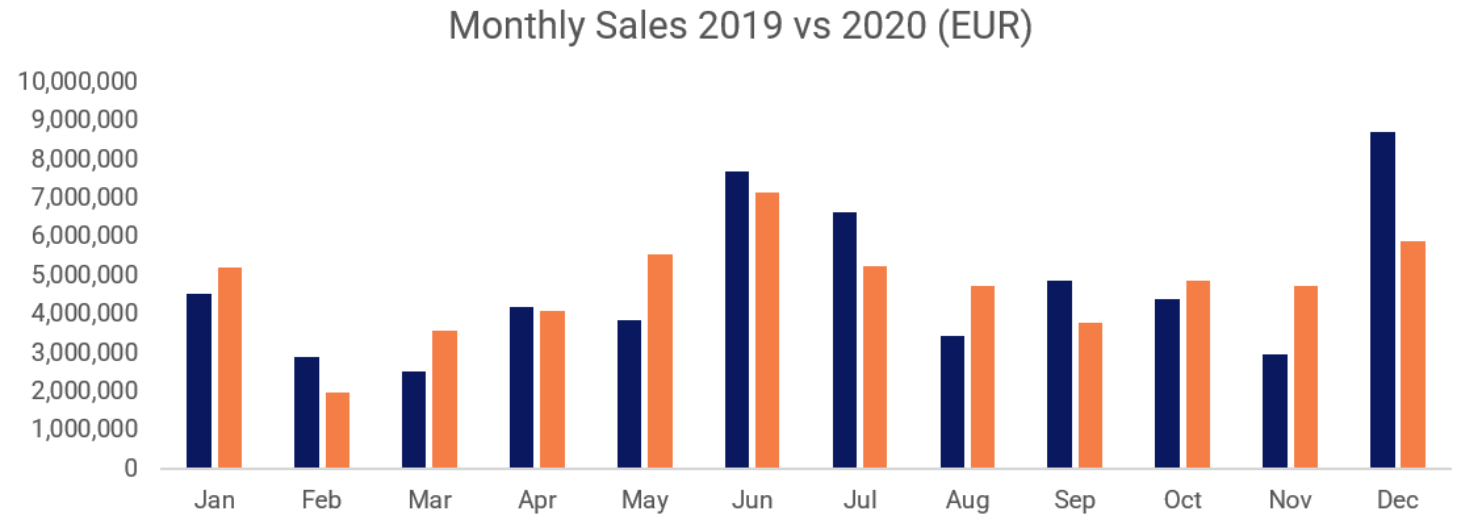
Step 1 — Insert the Chart

Pick the chart type that matches your data and question.

*Two series · Monthly time axis →
Grouped Bar Chart*

Raw first attempt — problems to fix:

- Y-axis shows raw numbers (10,000,000)
- Title gives no analytical signal
- Colours are default, no visual hierarchy



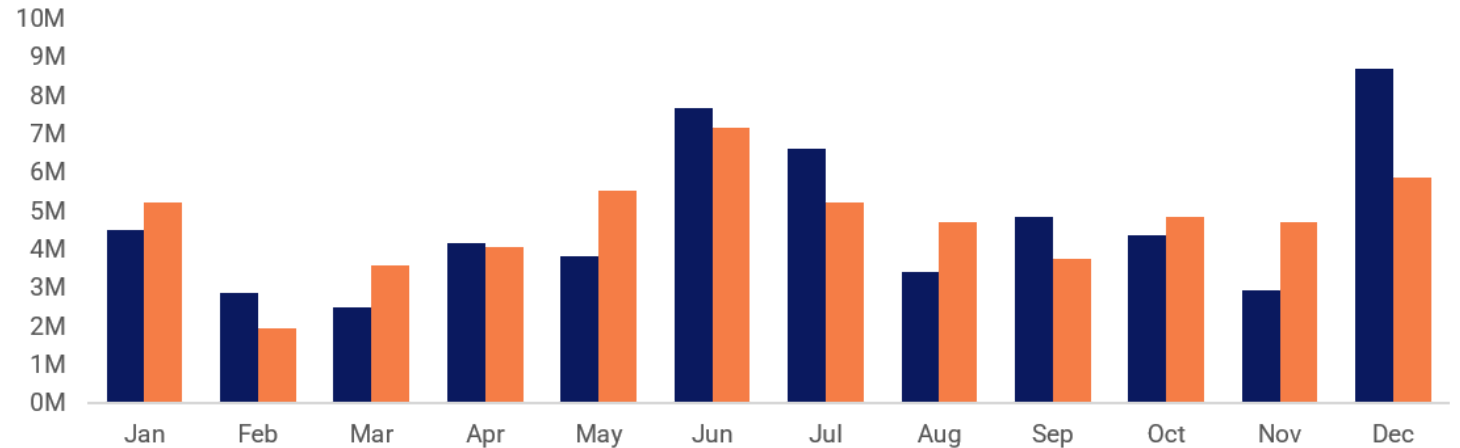
Step 2 – Format for Readability

Clean up anything that **slows the reader down**.

What changed:

- Y-axis: 10,000,000 → 10M (shorthand)
- Title: year labels **bolded** and **colour-coded** to match the bars
- Gridlines thinned; chart area decluttered

Monthly Sales **2019** vs **2020** (EUR)



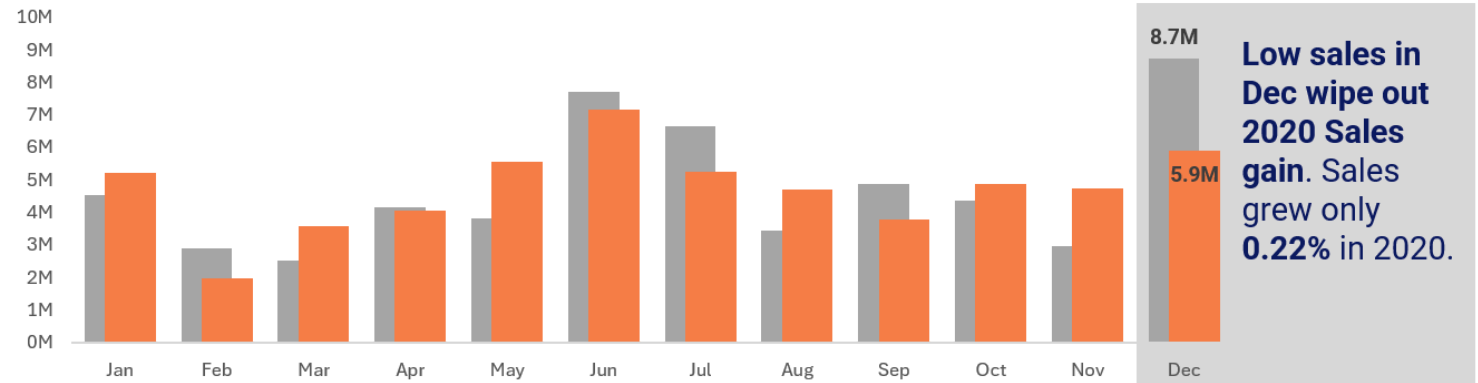
Step 3 – Focus on the Story

Decide what you want the reader to **notice first** – then make that visually obvious.

What changed:

- All non-December bars **greyed out**
- December data labels added:
8.7M / 5.9M
- Callout annotation states the insight in plain language

Monthly Sales 2019 vs 2020 (EUR)



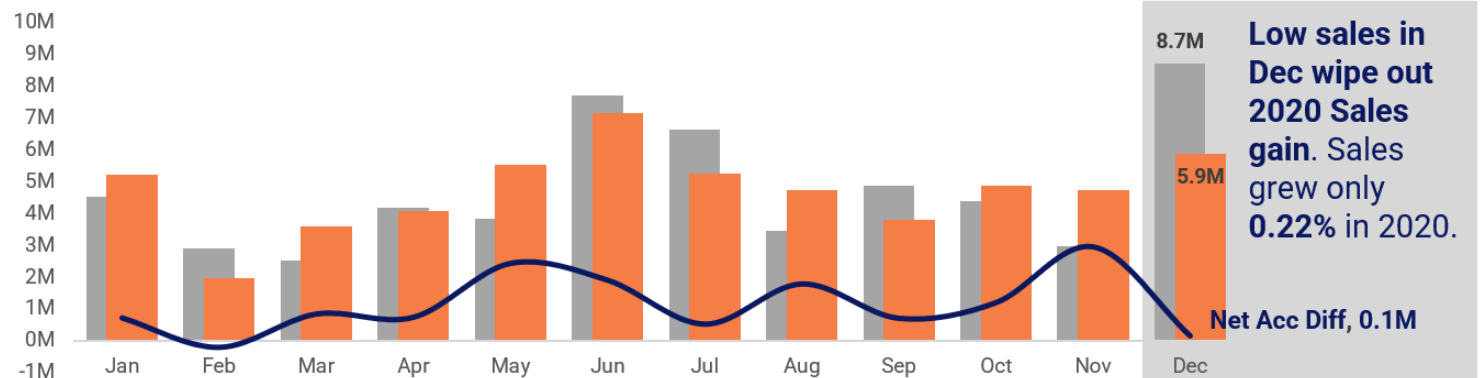
Step 4 – Add Context

Layer in a second data series to show **why** the story matters.

What changed:

- Net accumulated revenue difference added as an **overlaid line chart** (combo chart)
- December's sharp dip **confirms** the annotation

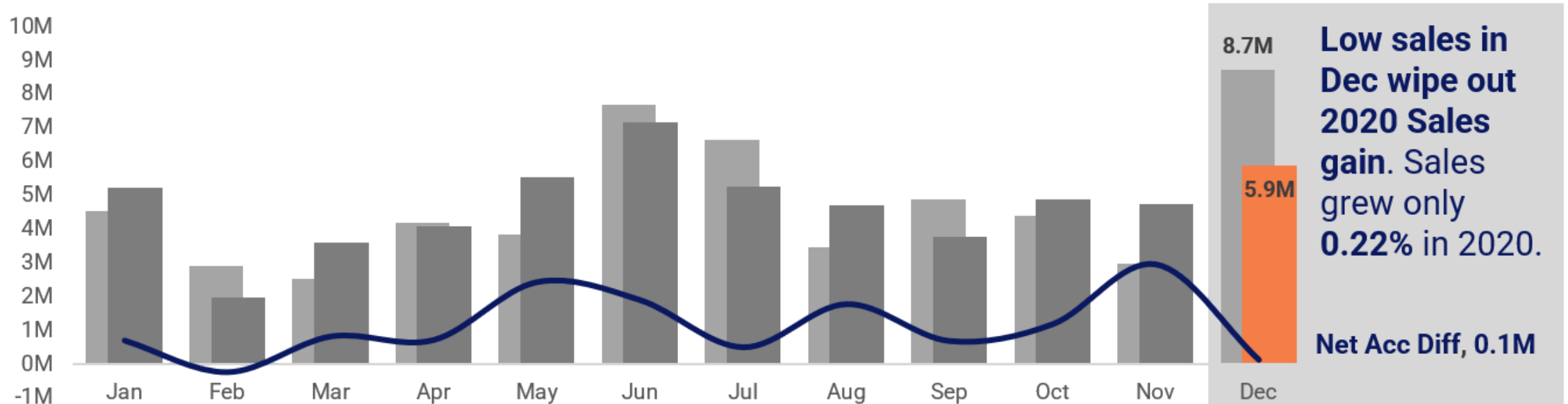
Monthly Sales 2019 vs 2020 (EUR)



Step 5 – The Finished Chart

LOSS OF CHRISTMAS SALES

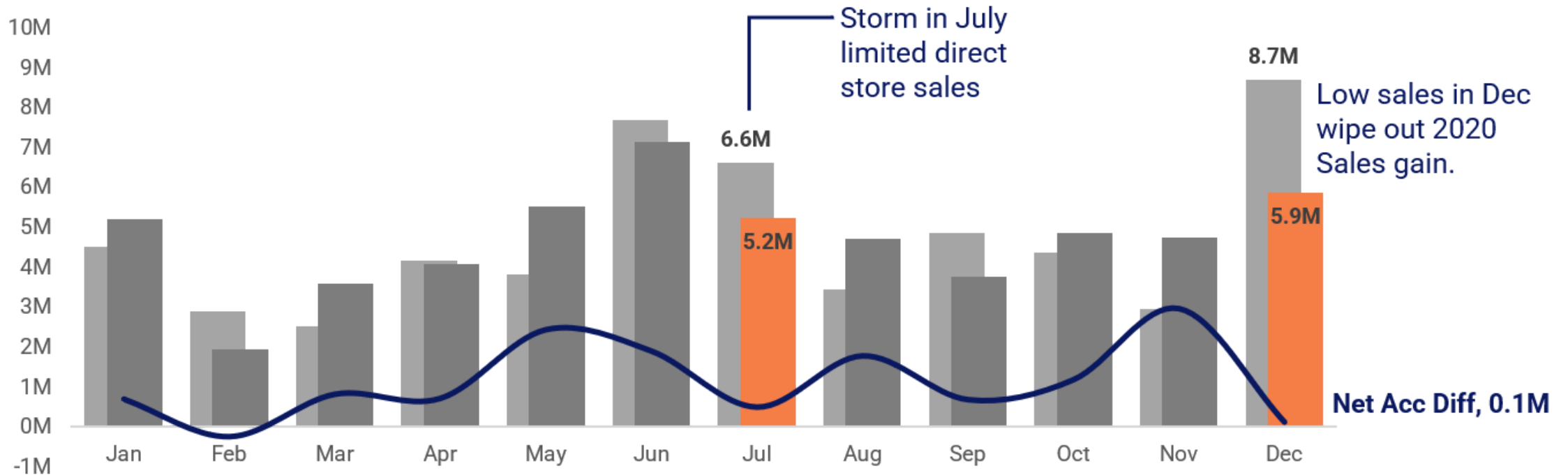
Monthly Sales 2019 vs 2020 (EUR)



Final refinement: make sure the **spotlight lands exactly where the story is.**

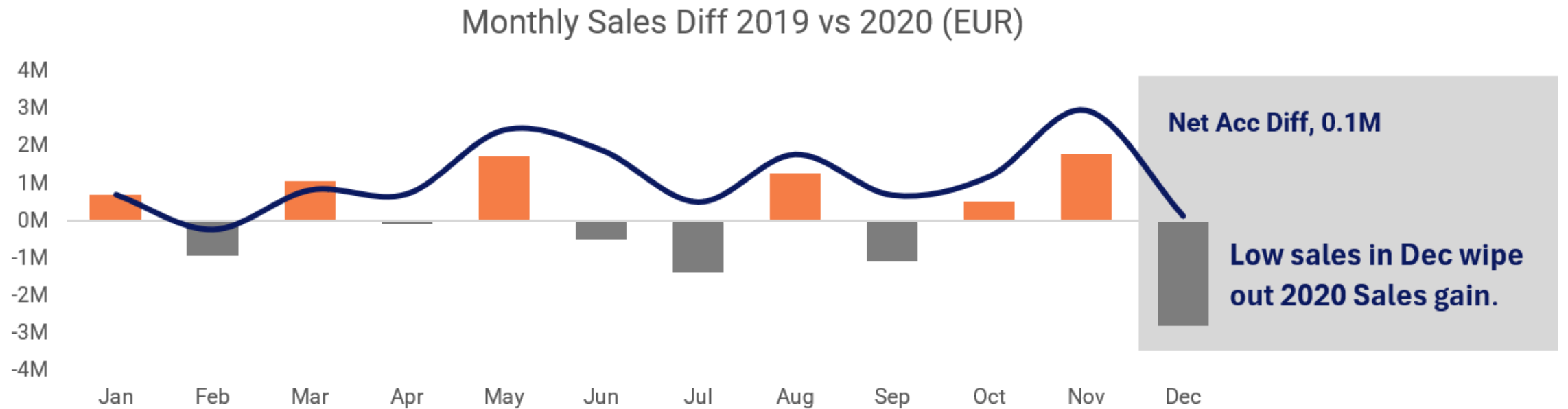
What if 2 Insights?

Monthly Sales 2019 vs 2020 (EUR)



A chart can carry **more than one story** — as long as each spotlight is clearly labelled and the two stories are directly connected.

Alternative Presentation

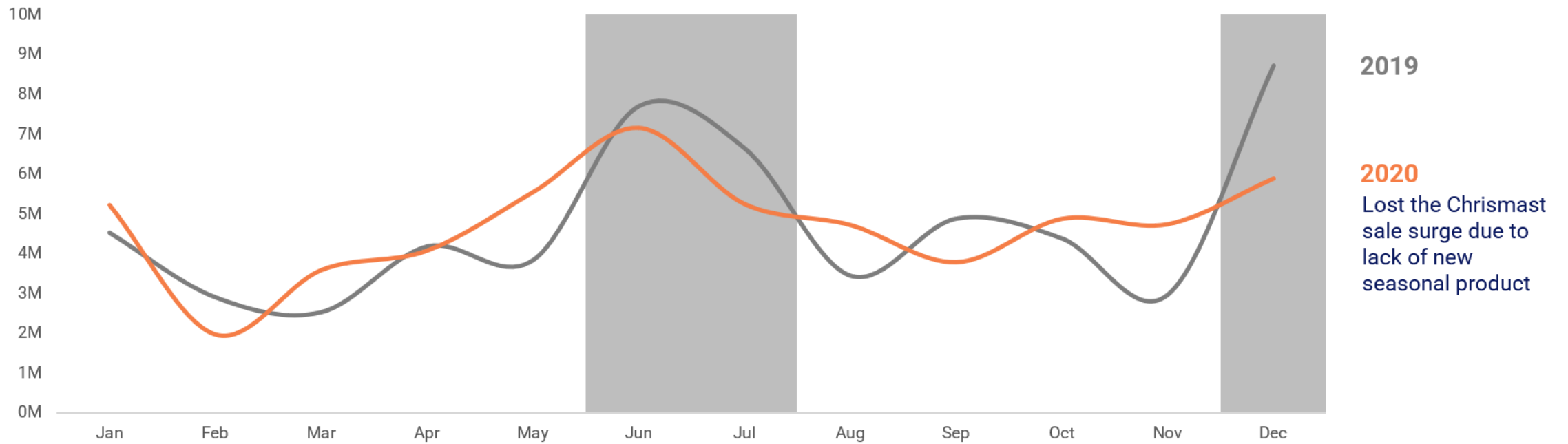


Plot the difference directly

Alternative Presentation 2

LOSS OF CHRISTMAS SALES

Monthly Sales 2019 vs 2020 (EUR)



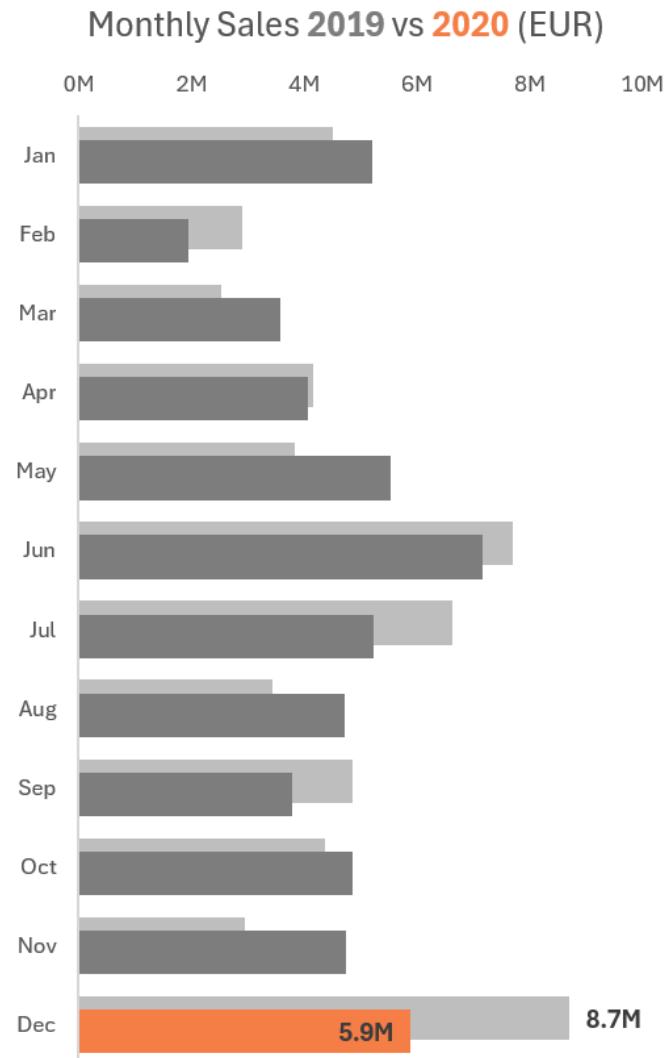
How about line chart?

Alternative Presentation 3 – ❌ Don't Do This

The same data — but this layout **breaks** a fundamental rule.

What is wrong:

- Our eyes read horizontal flow as time sequence — left to right
- A vertical time axis hides trends and makes the chart feel like a ranking, not a timeline



Ref

Day 1 — Key Takeaways

Core concepts

- **Data** = raw facts, no meaning alone
- **Information** = data with context
- **Insight** = information that drives action
- Visualisation accelerates the journey

Data types

Continuous · Discrete · Nominal · Ordinal · Temporal ·
Geospatial

Data type determines chart — not personal preference.

How to choose a chart

- Abela's tree — start with your message
- FT Visual Vocabulary — 9 purpose families
- Domain context — Finance, Sales, Science each have common patterns

Tools

- **Excel** — fast, offline, business standard
- **Python** — powerful, automated, reproducible

See You on Day 2

Finance Visualisation

Next session covers:

Trial Balance · P&L Statement · Balance Sheet · Waterfall Chart · Candlestick Chart

Questions before Day 2? Reach out anytime.

References

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